SONG FEI

Master and incoming PhD Student @ HKUST(GZ) · AIGC & Low Level Vision

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RESEARCH SUMMARY

Master student specializing in **Generative AI** and **Low Level Vision**. I build systems that define the **State-of-the-Art**:

- **SOTA Visual Frontiers:** Pioneered **LucidFlux** and **UltraFlux**, pushing visual generation quality beyond strong commercial baselines.

EDUCATION

The Hong Kong University of Science and Technology, Guangzhou — Guangzhou, China
Master Student in Robotics and Autonomous Systems (ROAS) — *Aug. 2024 – Present*

- Advisor: Prof. Lei Zhu and Prof. Jun Ma
- Research Focus: Text-to-Image Generation, Visual Restoration.

Northeastern University — Shenyang, China
B.Eng. in Industrial Intelligence — *Sep. 2020 – Jul. 2024*

RESEARCH EXPERIENCE

JD Retail — Beijing, China
TGT (Tech Genius Team) Research Intern — *Jun. 2026 – Present*

- **E-commerce Video Generation:** Worked on post-training for multi-reference image-conditioned and audio-driven video generation, improving product fidelity, temporal consistency, and audio-visual synchronization for retail content creation.

OPPO Research Institute — Shenzhen, China
Research Scientist Intern (Image Super-Resolution & Restoration) — *Dec. 2025 – Present*

- **Image Restoration & Super-Resolution:** Contributed to the research and development of advanced generative models for image super-resolution and degradation restoration, targeting real-world mobile photography scenarios. The proposed methods improved perceptual quality and detail reconstruction under low-light and compressed conditions.

The Hong Kong University of Science and Technology, Guangzhou — Guangzhou, China
Master Student — *Aug. 2024 – Present*

- **Visual Frontiers:** Served as a core contributor to **LucidFlux** (ICLR'26) and **UltraFlux** (CVPR'26). LucidFlux pushes the boundary of visual quality, surpassing strong commercial baselines (e.g., **Meitu SR**), and has been successfully launched as an **Official API on Fal.AI**.

SELECTED PUBLICATIONS

Summary: Published **8+** papers in top-tier conferences/journals (CVPR, ICCV, ICLR, MICCAI), including **4** first-author/co-first-author papers.

** Equal Contribution.*

First-Author & Co-First-Author Publications

[CVPR 2026 Highlight] UltraFlux: Data-Model Co-Design for High-quality Native 4K Text-to-Image Generation across Diverse Aspect Ratios
Tian Ye*, **Song Fei***, Lei Zhu.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[ICLR 2026] LucidFlux: Caption-Free Photo-Realistic Image Restoration via a Large-Scale Diffusion Transformer
Song Fei*, Tian Ye*, Lujia Wang, Lei Zhu.
International Conference on Learning Representations (ICLR), 2026.

[MICCAI 2026] S2DLight: Single Step Diffusion Network via Degradation-Aware Adaptation for Surgical Endoscopic Image Low-Light Enhancement
Song Fei, Guanyi Wang, Tian Ye, Sixiang Chen, Jianyu Lai, Lei Zhu.
International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.

[NeurIPS 2026] LucidNFT: LR-Anchored Multi-Reward Preference Optimization for Flow-Based Real-World Super-Resolution
Song Fei*, Tian Ye*, Sixiang Chen, Zhaohu Xing, Jianyu Lai, Lei Zhu.
Neural Information Processing Systems (NeurIPS), 2026.

Co-Authored Publications

[MICCAI 2026] MoSE: Mixture-of-Scale-Experts for Medical Image Restoration
Xingyu Li, Zhaohu Xing, Zekai Liu, Shuaibo Li, Hongqiu Wang, **Song Fei**, Lei Zhu.
International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.

[ICLR 2026] PosterCraft: Rethinking High-Quality Aesthetic Poster Generation in a Unified Framework
SiXiang Chen*, Jianyu Lai*, Jialin Gao*, Tian Ye, Haoyu Chen, Hengyu Shi, Shitong Shao, Yunlong Lin, **Song Fei**, Zhaohu Xing, Yeying Jin, Junfeng Luo, Xiaoming Wei, Lei Zhu.
International Conference on Learning Representations (ICLR), 2026.

[CVPR 2025] SnowMaster: Comprehensive Real-world Image Desnowing via MLLM
Jianyu Lai, Sixiang Chen, Yunlong Lin, Tian Ye, Yun Liu, **Song Fei**, Zhaohu Xing, Hongtao Wu, Weiming Wang, Lei Zhu.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

[ICCV 2025] GenHaze: Pioneering Controllable One-Step Realistic Haze Generation for Real-World Dehazing
Sixiang Chen, Tian Ye, Yunlong Lin, Yeying Jin, Yijun Yang, Haoyu Chen, Jianyu Lai, **Song Fei**, Zhaohu Xing, Fugee Tsung, Lei Zhu.
International Conference on Computer Vision (ICCV), 2025.

Honors & Academic Service

Honors & Awards
- Postgraduate Scholarship of HKUST(GZ), 2024

Academic Service
- **Conference Reviewer:** AAAI.